9.26.

a)

$k = \omega^2\mu = (2\pi\nu)^2\mu = (2\pi c\tilde{\nu})^2\mu$

$^{35}Cl_2$

$\tilde{\nu} = 560\ \mathrm{cm}^{-1}$

$$\mu = \frac{m_{Cl}m_{Cl}}{m_{Cl} + m_{Cl}} \approx \frac{35 \cdot 35}{35 + 35} 1.66 \cdot 10^{-27}\mathrm{kg} \approx 2.905 \cdot 10^{-26}\mathrm{kg}$$

$$k = (2 \cdot 3.14 \cdot 3 \cdot 10^{10}\mathrm{cm} \cdot \mathrm{s}^{-1} \cdot 560\ \mathrm{cm}^{-1})^2 2.905 \cdot 10^{-26}\mathrm{kg} \approx 323\ \mathrm{Nm}^{-1}$$

$^{39}K^{35}Cl$

$\tilde{\nu} = 281\ \mathrm{cm}^{-1}$

$$\mu = \frac{m_K m_{Cl}}{m_K + m_{Cl}} \approx \frac{39 \cdot 35}{39 + 35} 1.66 \cdot 10^{-27}\mathrm{kg} \approx 3.06 \cdot 10^{-26}\mathrm{kg}$$

$$k = (2 \cdot 3.14 \cdot 3 \cdot 10^{10}\mathrm{cm} \cdot \mathrm{s}^{-1} \cdot 281\ \mathrm{cm}^{-1})^2 3.06 \cdot 10^{-26}\mathrm{kg} \approx 85.8\ \mathrm{Nm}^{-1}$$

$^{1}H_2$

$\tilde{\nu} = 4401\ \mathrm{cm}^{-1}$

$$\mu = \frac{m_H m_H}{m_H + m_H} \approx \frac{1 \cdot 1}{1 + 1} 1.66 \cdot 10^{-27}\mathrm{kg} \approx 8.3 \cdot 10^{-28}\mathrm{kg}$$

$$k = (2 \cdot 3.14 \cdot 3 \cdot 10^{10}\mathrm{cm} \cdot \mathrm{s}^{-1} \cdot 4401\ \mathrm{cm}^{-1})^2 8.3 \cdot 10^{-28}\mathrm{kg} \approx 571\ \mathrm{Nm}^{-1}$$

b)

$$\tilde{\nu} \sim \sqrt{\frac{1}{\mu}}$$

$^{35}Cl_2$

$\tilde{\nu}_1 = 560\ \mathrm{cm}^{-1}$

$$\mu_1 = \frac{m_{Cl} m_{Cl}}{m_{Cl} + m_{Cl}} = \frac{35 \cdot 35}{35 + 35} = \frac{35}{2}\ \mathrm{Da}$$

$^{37}Cl_2$

$$\mu_2 = \frac{m_{Cl} m_{Cl}}{m_{Cl} + m_{Cl}} = \frac{37 \cdot 37}{37 + 37} = \frac{37}{2}\ \mathrm{Da}$$

$$\tilde{\nu}_2 = \tilde{\nu}_1 \sqrt{\frac{\mu_1}{\mu_2}} \approx 560 \sqrt{\frac{35}{37}} \approx 545\ \mathrm{cm}^{-1}$$

9.28.

The Hamiltonian for the harmonic oscillator:

$$\hat{H} = \hat{T} + \hat{V} = -\frac{\hbar^2}{2m}\frac{d^2}{dx^2} + \frac{1}{2}m\omega^2 x^2$$

Eigenfunction of the first excited state:

$$\Psi_1(x) = \left(\frac{4\alpha^3}{\pi}\right)^{1/4} xe^{-\frac{\alpha x^2}{2}}$$

Substituting the eigenfunction into Schrödinger equation:

$$-\frac{\hbar^2}{2m}\frac{d^2}{dx^2}\left(xe^{-\frac{\alpha x^2}{2}}\right) + \frac{1}{2}m\omega^2 x^2\left(xe^{-\frac{\alpha x^2}{2}}\right) = E_1 xe^{-\frac{\alpha x^2}{2}}$$

Take second derivative:

$$\frac{d^2}{dx^2}\left(xe^{-\frac{\alpha x^2}{2}}\right) = \frac{d}{dx}\left(e^{-\frac{\alpha x^2}{2}} - \alpha x^2 e^{-\frac{\alpha x^2}{2}}\right) = -\alpha xe^{-\frac{\alpha x^2}{2}} - 2\alpha xe^{-\frac{\alpha x^2}{2}} + \alpha^2 x^3 e^{-\frac{\alpha x^2}{2}} =$$

$$= (-3\alpha + \alpha^2 x^2)xe^{-\frac{\alpha x^2}{2}} = (-\frac{3m\omega}{\hbar} + \frac{m^2\omega^2}{\hbar^2}x^2)xe^{-\frac{\alpha x^2}{2}}$$

Substituting in Schrödinger equation and canceling $xe^{-\frac{\alpha x^2}{2}}$

$$\frac{3\hbar\omega}{2} - \frac{1}{2}m\omega^2x^2 + \frac{1}{2}m\omega^2x^2 = E_1$$

Or

$$E_1 = \frac{3}{2}\hbar\omega$$

9.29.

$$k = \omega^2\mu = (2\pi\nu)^2\mu = (2\pi c\tilde{\nu})^2\mu$$

$$\tilde{\nu} = \frac{1}{2\pi c}\sqrt{\frac{k}{\mu}} \approx \frac{1}{2 \cdot 3.14 \cdot 3 \cdot 10^{10}}\sqrt{\frac{317}{1.66 \cdot 10^{-27}}} \approx 2318\ \mathrm{cm}^{-1}$$

If H is replaced by D:

$$\tilde{\nu}_D = \tilde{\nu}_H\sqrt{\frac{m_H}{m_D}} \approx 1639\ \mathrm{cm}^{-1}$$

9.31.

a)
$^{12}C^{16}O$

$\tilde{\nu} = 2170\ \mathrm{cm}^{-1}$

$$\mu = \frac{m_C m_O}{m_C + m_O} \approx \frac{12 \cdot 16}{12 + 16} 1.66 \cdot 10^{-27}\mathrm{kg} \approx 1.14 \cdot 10^{-26}\mathrm{kg}$$

$$k = (2 \cdot 3.14 \cdot 3 \cdot 10^{10}\mathrm{cm} \cdot \mathrm{s}^{-1} \cdot 2170\ \mathrm{cm}^{-1})^2 1.14 \cdot 10^{-26}\mathrm{kg} \approx 1900\ \mathrm{Nm}^{-1}$$

b)

$$\Delta x = \left(\frac{1}{2\alpha}\right)^{1/2} = \left(\frac{\hbar^2}{4k\mu}\right)^{1/4} \approx \left(\frac{(1.05 \cdot 10^{-34})^2}{4 \cdot 1900 \cdot 1.14 \cdot 10^{-26}}\right)^{1/4} \approx 3.4\ \mathrm{pm}$$

c)

$$\Delta p = \left(\frac{\alpha\hbar^2}{2}\right)^{1/2} = \left(\frac{k\mu\hbar^2}{4}\right)^{1/4} \approx \left(\frac{1900 \cdot 1.14 \cdot 10^{-26}(1.05 \cdot 10^{-34})^2}{4}\right)^{1/4} \approx$$
$$\approx 1.56 \cdot 10^{-23}\ \mathrm{kg\ m\ s}^{-1}$$

d)

$$\Delta x \Delta p \approx 3.4 \cdot 10^{-12} \cdot 1.56 \cdot 10^{-23} \approx 5.3 \cdot 10^{-35}\mathrm{J\ s} \approx \frac{\hbar}{2}$$

9.32.

$$\Delta E = E_2 - E_1 = \frac{\hbar^2}{2I}(2 \cdot 3 - 1 \cdot 2) = \frac{2\hbar^2}{\mu R_e^2}$$

$$\nu = \frac{\Delta E}{h} = \frac{h}{2\pi^2 I} \approx \frac{6.63 \cdot 10^{-34}}{2\pi^2 2.64 \cdot 10^{-47}} \approx 1.27 \cdot 10^{12}\ \mathrm{s}^{-1}$$

$$\tilde{\nu} = \frac{\nu}{c} \approx \frac{1.27 \cdot 10^{12}}{3 \cdot 10^{10}} \approx 42.3\ \mathrm{cm}^{-1}$$

9.34.

$^{23}Na^{35}Cl$

$$\mu = \frac{m_{Na}m_{Cl}}{m_{Na} + m_{Cl}} \approx \frac{23 \cdot 35}{23 + 35} 1.66 \cdot 10^{-27} \approx 2.3 \cdot 10^{-26}\text{kg}$$

$$I = \mu R_e^2 \approx 2.3 \cdot 10^{-26}(2.36 \cdot 10^{-10})^2 \approx 1.28 \cdot 10^{-45}\text{kg m}^2$$

$$E_J = \frac{\hbar^2}{2I} J(J+1)$$

$$E_1 = \frac{\hbar^2}{I} \approx \frac{(1.05 \cdot 10^{-34})^2}{1.28 \cdot 10^{-45}} \approx 8.6 \cdot 10^{-24}\text{ J}$$

$$E_2 = \frac{3\hbar^2}{I} \approx \frac{3(1.05 \cdot 10^{-34})^2}{1.28 \cdot 10^{-45}} \approx 2.6 \cdot 10^{-23}\text{ J}$$

9.36.

$^{12}C^{16}O$
$R_e = 112.8\ \text{pm}$

a)

$$\mu = \frac{m_C m_O}{m_C + m_O} \approx \frac{12 \cdot 16}{12 + 16} 1.66 \cdot 10^{-27}\text{kg} \approx 1.14 \cdot 10^{-26}\text{kg}$$

b)

$$I = \mu R_e^2 \approx 1.14 \cdot 10^{-26}(1.128 \cdot 10^{-10})^2 \approx 1.45 \cdot 10^{-46}\text{kg m}^2$$

c)

$$\Delta E = E_1 - E_0 = \frac{\hbar^2}{2I}(1 \cdot 2 - 0 \cdot 1) = \frac{\hbar^2}{I}$$

$$\lambda = \frac{hc}{\Delta E} = \frac{2\pi c I}{\hbar} \approx \frac{2 \cdot 3.14 \cdot 3 \cdot 10^8 \cdot 1.45 \cdot 10^{-46}}{1.05 \cdot 10^{-34}} \approx 2.6 \cdot 10^{-3}\ \text{m}$$

9.54.

$$E_{v_x v_y v_z} = \frac{\hbar}{\sqrt{\mu}}\left((v_x + 1/2)\sqrt{k_x} + (v_y + 1/2)\sqrt{k_y} + (v_z + 1/2)\sqrt{k_z}\right)$$

$$E_{000} = \frac{\hbar}{2\sqrt{\mu}}\left(\sqrt{k_x} + \sqrt{k_y} + \sqrt{k_z}\right)$$

9.55.

$$E_{v_x v_y v_z} = \hbar\sqrt{\frac{k}{\mu}}\left(v_x + v_y + v_z + \frac{3}{2}\right)$$

ZPE:

$$E_{000} = \frac{3}{2}\hbar\sqrt{\frac{k}{\mu}}$$

Non-degenerate

First excited state:

$$E_{100} = \frac{5}{2}\hbar\sqrt{\frac{k}{\mu}}$$

Degeneracy: 3 (100, 010, 001)

Second excited state:

$$E_{200} = \frac{7}{2}\hbar\sqrt{\frac{k}{\mu}}$$

Degeneracy: 6 (200, 020, 002, 110, 101, 011)

Third excited state:

$$E_{300} = \frac{9}{2}\hbar\sqrt{\frac{k}{\mu}}$$

Degeneracy: 10 (300, 030, 003, 210, 201, 021, 120, 102, 012, 111)

Fourth excited state:

$$E_{400} = \frac{11}{2}\hbar\sqrt{\frac{k}{\mu}}$$

Degeneracy: 15 (400, 310, 301, 211, 220, 202, 130, 103, 121, 112, 040, 004, 031, 013, 022)

Fifth excited state:

$$E_{500} = \frac{13}{2}\hbar\sqrt{\frac{k}{\mu}}$$

Degeneracy: 21 $(D_n = D_{n-1} + n + 1)$

Sixth excited state:

$$E_{600} = \frac{15}{2}\hbar\sqrt{\frac{k}{\mu}}$$

Degeneracy: 28 $(D_n = D_{n-1} + n + 1)$

Seventh excited state:

$$E_{700} = \frac{17}{2}\hbar\sqrt{\frac{k}{\mu}}$$

Degeneracy: 36 $(D_n = D_{n-1} + n + 1)$

Eights excited state:

$$E_{800} = \frac{19}{2}\hbar\sqrt{\frac{k}{\mu}}$$

Degeneracy: 45 $(D_n = D_{n-1} + n + 1)$

Ninth excited state:

$$E_{900} = \frac{21}{2}\hbar\sqrt{\frac{k}{\mu}}$$

Degeneracy: 55 $(D_n = D_{n-1} + n + 1)$

9.59.

$^{14}N^{16}O$

$R_e = 115.1\ \text{pm}$

$$\mu = \frac{m_N m_O}{m_N + m_O} \approx \frac{14 \cdot 16}{14 + 16} 1.66 \cdot 10^{-27}\text{kg} \approx 1.24 \cdot 10^{-26}\text{kg}$$

$$I = \mu R_e^2 \approx 1.24 \cdot 10^{-26}(1.151 \cdot 10^{-10})^2 \approx 1.64 \cdot 10^{-46}\text{kg m}^2$$

$$E_0 = 0$$

$$E_1 = \frac{\hbar^2}{2I} 1(1+1) = \frac{\hbar^2}{I} \approx \frac{(1.05 \cdot 10^{-34})^2}{1.64 \cdot 10^{-46}} \approx 6.7 \cdot 10^{-23}\ \text{J}$$

$$E_2 = \frac{\hbar^2}{2I} 2(2+1) = \frac{3\hbar^2}{I} \approx 2.0 \cdot 10^{-22}\ \text{J}$$

Bonus Problem 1. (15 pts) **Spectroscopic selection rules for a harmonic oscillator.**

In this problem, we will analyze the spectroscopic selection rules for a harmonic oscillator and then consider the effect of anharmonicity.

In spectroscopy, transition probability P_{nm} between quantum states $|\psi_m\rangle$ and $|\psi_n\rangle$ is determined by the *transition dipole moment*

$$\mu_{nm} = \langle \psi_n | \mu(x) | \psi_m \rangle$$

The transition probability is $P_{nm} \propto |\mu_{nm}|^2$. This probability is often called the oscillator strength. The dipole moment of a molecule is a function of its vibrational coordinate x. For small vibrations, the dipole moment can be expanded into Taylor series

$$\mu(x) = \mu_0 + \frac{\partial \mu}{\partial x} x + \frac{1}{2!} \frac{\partial^2 \mu}{\partial x^2} x^2 + \cdots$$

Using Dirac notation and the raising and lowering operators, we will evaluate the integrals $\langle \psi_n | \mu(x) | \psi_m \rangle$ for the harmonic oscillator to derive the *selection rules* for

vibrational spectroscopy: if $\langle \psi_n | \mu(x) | \psi_m \rangle = 0$, the $n \leftarrow m$ transition is called *forbidden*, and if $\langle \psi_n | \mu(x) | \psi_m \rangle \neq 0$, the transition is *allowed*.

a) (7 pts) In the first approximation you can use the first two terms of the Taylor series expansion for the dipole moment: $\mu(x) = \mu_0 + \frac{\partial \mu}{\partial x} x$
Calculate the transition dipole moment and derive the general selection rules for vibrational spectroscopy in this approximation. These transitions are called *fundamental* transitions.

$$\mu_{nm} = \langle \psi_n | \mu(x) | \psi_m \rangle \approx \left\langle \psi_n \middle| \left(\mu_0 + \frac{\partial \mu}{\partial x} \hat{x} \right) \middle| \psi_m \right\rangle = \langle \psi_n | \mu_0 | \psi_m \rangle + \left\langle \psi_n \middle| \frac{\partial \mu}{\partial x} \hat{x} \middle| \psi_m \right\rangle =$$
$$= \mu_0 \langle \psi_n | \psi_m \rangle + \frac{\partial \mu}{\partial x} \langle \psi_n | \hat{x} | \psi_m \rangle$$

$$\langle \psi_n | \psi_m \rangle = 0 \quad \text{for n} \neq \text{m}$$

$$\hat{x} = \sqrt{\frac{\hbar}{2\mu\omega}} (\hat{a} + \hat{a}^\dagger)$$

$$\hat{a} | \psi_m \rangle = \sqrt{m} | \psi_{m-1} \rangle$$

$$\hat{a}^\dagger | \psi_m \rangle = \sqrt{m+1} | \psi_{m+1} \rangle$$

$$\langle \psi_n | \hat{x} | \psi_m \rangle = \sqrt{\frac{\hbar}{2\mu\omega}} \langle \psi_n | (\hat{a} + \hat{a}^\dagger) | \psi_m \rangle =$$
$$= \sqrt{\frac{\hbar}{2\mu\omega}} \left(\sqrt{m} \langle \psi_n | \psi_{m-1} \rangle + \sqrt{m+1} \langle \psi_n | \psi_{m+1} \rangle \right)$$

Transition is allowed, when $n = m \pm 1$ and $\frac{\partial \mu}{\partial x} \neq 0$

b) (3 pts) Are the 1←0, 2←1, 3←2, etc. transition expected to have the same oscillator strength?

$P_{nn-1} \propto |\langle\psi_n|\hat{x}|\psi_{n-1}\rangle|^2 \propto n$

Transition 2←1 is two times stronger than 1←0; transition 3←2 is three times stronger than 1←0

c) (5 pts) How do the transition selection rules change if we take into consideration the third term for the dipole moment?

$$\left\langle\psi_n\left|\frac{1}{2!}\frac{\partial^2\mu}{\partial x^2}\hat{x}^2\right|\psi_m\right\rangle = \frac{\hbar}{4\mu\omega}\frac{\partial^2\mu}{\partial x^2}\langle\psi_n|(\hat{a}+\hat{a}^\dagger)^2|\psi_m\rangle =$$
$$= \frac{\hbar}{4\mu\omega}\frac{\partial^2\mu}{\partial x^2}\left\langle\psi_n\left|\hat{a}^2+\hat{a}\hat{a}^\dagger+\hat{a}^\dagger\hat{a}+\hat{a}^{\dagger 2}\right|\psi_m\right\rangle$$

$$\langle\psi_n|\hat{a}^2|\psi_m\rangle = \sqrt{m}\sqrt{m-1}\langle\psi_n|\psi_{m-2}\rangle$$

$$\left\langle\psi_n\left|\hat{a}^{\dagger 2}\right|\psi_m\right\rangle = \sqrt{m+1}\sqrt{m+2}\langle\psi_n|\psi_{m+2}\rangle$$

$$\langle\psi_n|\hat{a}\hat{a}^\dagger|\psi_m\rangle = (m+1)\langle\psi_n|\psi_m\rangle = 0 \text{ for n} \neq \text{m}$$
$$\langle\psi_n|\hat{a}^\dagger\hat{a}|\psi_m\rangle = m\langle\psi_n|\psi_m\rangle = 0 \text{ for n} \neq \text{m}$$

Additional selection rule is $n = m \pm 2$ and $\frac{\partial^2\mu}{\partial x^2} \neq 0$